BRAC University

Department of Computer Science and Engineering

CSE422: Artificial Intelligence

Lab Project Report

Hotel Booking Reservation Status Classification

Using Machine Learning Techniques

Submitted by

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1 Introduction

Hotel reservations generate large volumes of operational data that can be used to understand whether a booking will be completed, canceled, or recorded as a no-show. Uncertainty in reservation outcomes affects room availability, revenue planning, staffing, overbooking policies, and customer-service decisions. This project develops a machine-learning workflow for predicting reservation status from booking characteristics while carefully avoiding variables that directly disclose the final outcome.

The task is formulated as a three-class classification problem with the target classes Canceled, Check-Out, and No-Show. Four supervised algorithms - K-Nearest Neighbors, Logistic Regression, Gaussian NaiveBayes, and a multilayer neural network - are trained and compared. The same dataset also examined an unsupervised problem using K-Means clustering to identify natural booking profiles without using reservation-status labels.

The main motivation is to determine whether patterns such as lead time, deposit type, market segment, customer type, special requests, previous cancellations, and booking characteristics can support useful prediction. Because the No-Show class is extremely rare, the study considers not only accuracy but also balanced accuracy, macro-averaged measures, class-specific results, confusion matrices, and multiclass ROC-AUC.

2 Dataset Description

2.1 Dataset Overview

The original hotel-booking dataset contains 119,390 observations and 32 columns. Twenty columns are numerical and twelve are categorical before preprocessing. Numerical examples include lead_time, adr, stays_in_weekend_nights, stays_in_week_nights, adults, children, previous_cancellations, booking_changes, required_car_parking_spaces, and total_of_special_requests. Categorical examplesinclude hotel, arrival_date_month, meal, country, market_segment, distribution_channel, roomtype, deposit_type, agent, customer_type, and reservation_status.

Table 1: Dataset Overview

Property	Value
Original observations	119,390
Original columns	32
Numerical columns	20
Categorical columns	12
Target	reservation_status
Target classes	Canceled, Check-Out, No-Show

2.1.1 Classification or Regression Problem?

This is a multiclass classification problem because the output is one of three discrete labels. It is not regression because the model does not estimate a continuous numerical quantity.

2.1.2 Need for Categorical Encoding

Encoding is required because the selected machine-learning algorithms operate on numerical arrays and cannot directly interpret text values such as City Hotel, Portugal, Online TA, or No Deposit. The model pipeline applies one-hot encoding to categorical input features and label encoding to the target. Unknown categories in test data are ignored safely by the encoder.

2.2 Correlation Analysis

A complete exploratory correlation matrix was produced after temporarily label-encoding categorical columns. This made it possible to include all 32 input and output columns in a seaborn heatmap. The encoding was used only for exploration; the training pipeline uses one-hot encoding because integer labels for nominal categories do not represent a natural order.

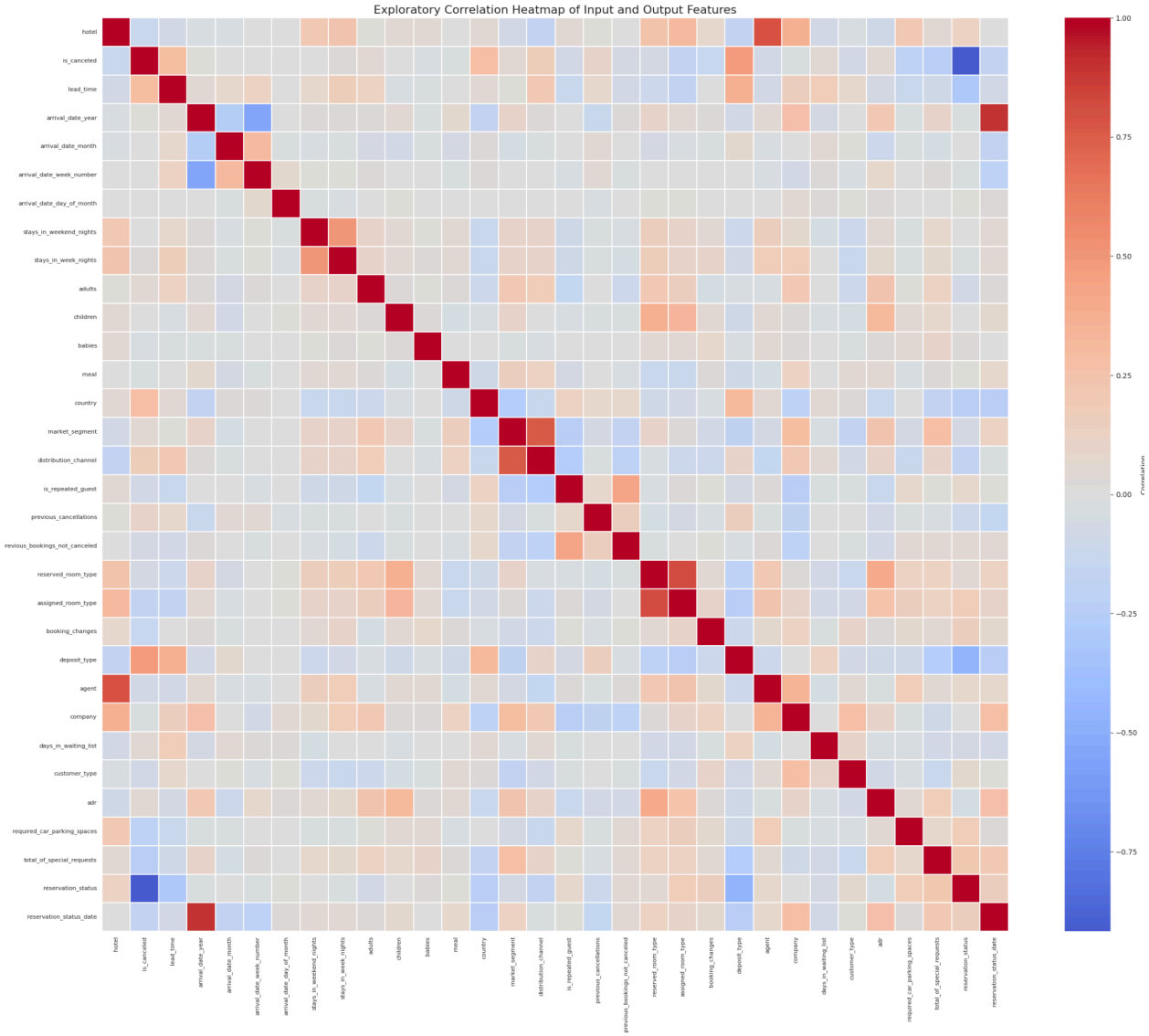


Figure 1: Exploratory Correlation Heatmap of All Input and Output Features

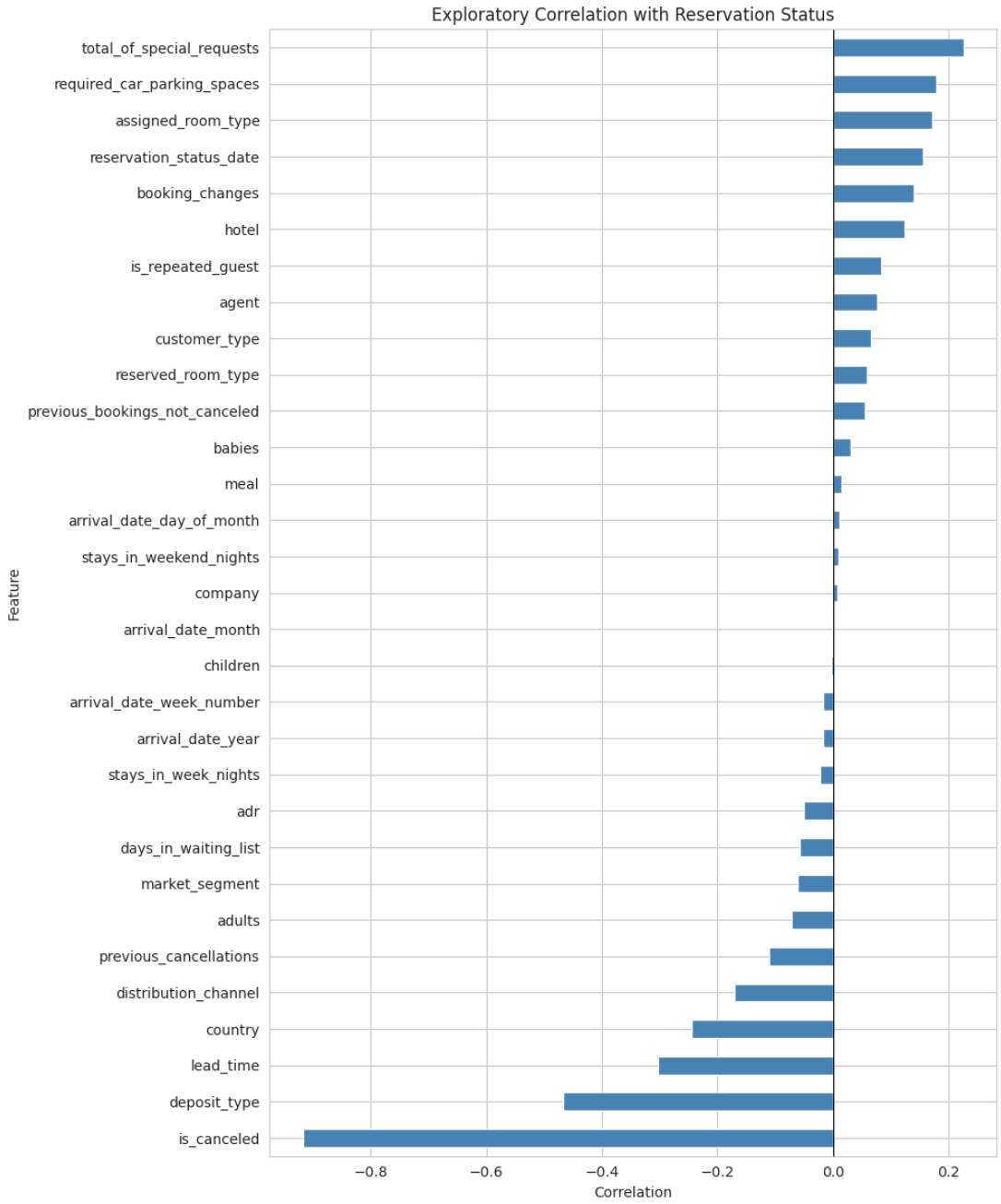


Figure 2: Exploratory Correlation of Features with Reservation Status

The strongest apparent relationship is between is_canceled and reservation_status (approximately-0.92). This is expected because cancellation status directly defines much of the target and therefore constitutes target leakage. Deposit type and lead time also show notable exploratory relationships. Special requests, required parking spaces, distribution channel, booking changes, and hotel type exhibit smaller associations. Since several variables are nominal, the exact correlation magnitudes involving encoded categories should be interpreted as exploratory signals rather than ordinal effects.

2.3 Imbalanced Dataset Analysis

Table 2: Original Class Distribution

Reservation status	Count	Percentage
Check-Out	75,166	62.96%
Canceled	43,017	36.03%
No-Show	1,207	1.01%

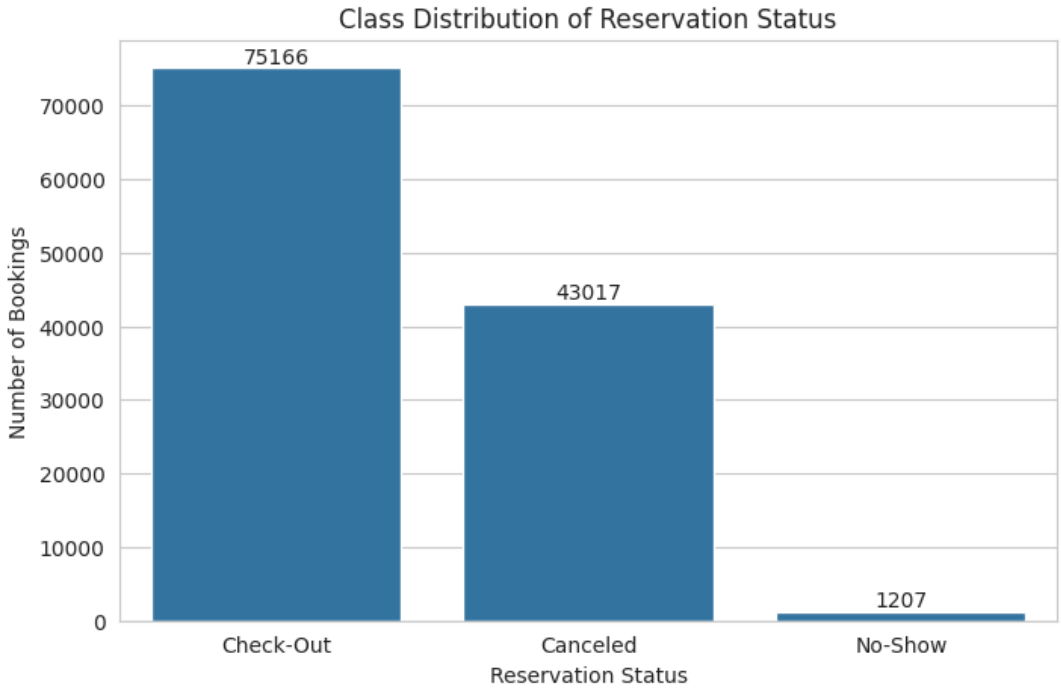


Figure 3: Number of Bookings in Each Reservation-Status Class

The target is severely imbalanced. Check-Out represents nearly two thirds of the records, whereas No-Show accounts for only about one percent. Consequently, overall accuracy and weighted metrics may appear strong even when a model detects very few no-shows. Macro F1, balanced accuracy, per-class recall, and confusion matrices are therefore emphasized during model comparison.

2.4 Exploratory Data Analysis

Exploratory analysis was conducted to identify relationships between reservation outcomes and operational booking variables. The following plots compare status across hotel type, arrival month, market segment, deposit type, customer type, lead time, average daily rate, and special requests.



Figure 4: Reservation Status by Hotel Type

City hotels contribute a larger number of bookings and cancellations than resort hotels. This suggests that hotel type carries useful information, although the effect must be considered together with volume differences.

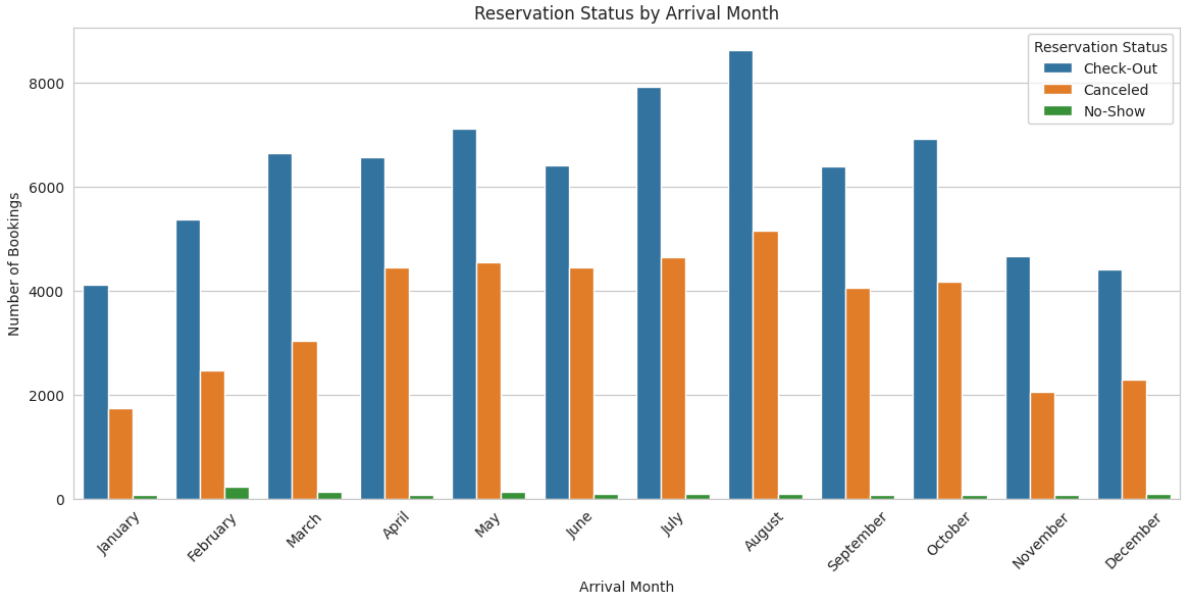


Figure 5: Reservation Status by Arrival Month

Reservation activity changes throughout the year, and cancellation counts tend to rise during high-volume months. Month therefore captures seasonality that may affect demand and cancellation behavior.

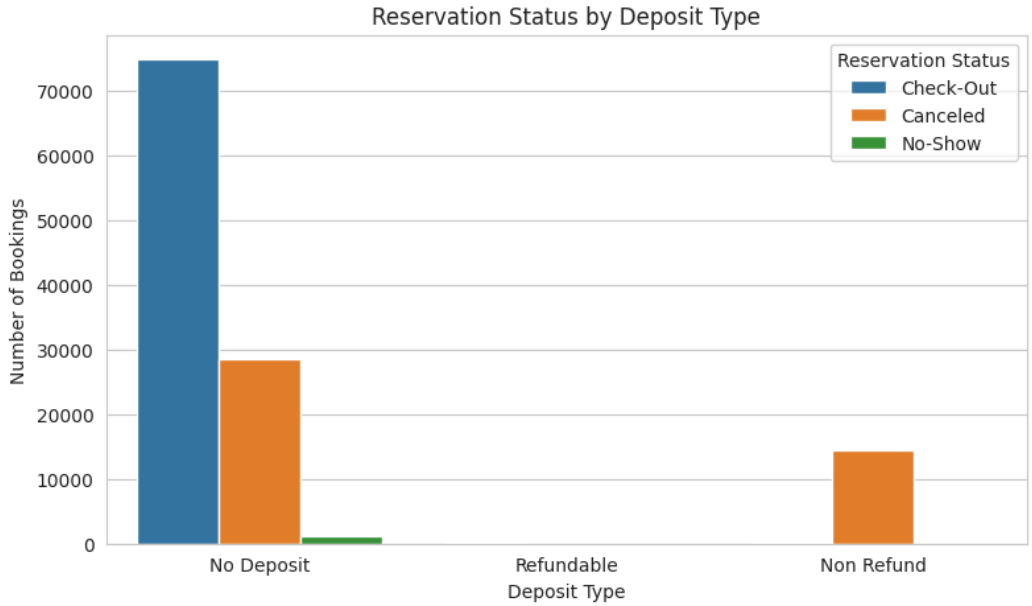


Figure 6: Reservation Status by Deposit Type

Deposit type shows a strong visual relationship with reservation outcome. Non-refundable arrangements behave differently from no-deposit bookings, making this one of the most informative categorical predictors retained during feature selection.

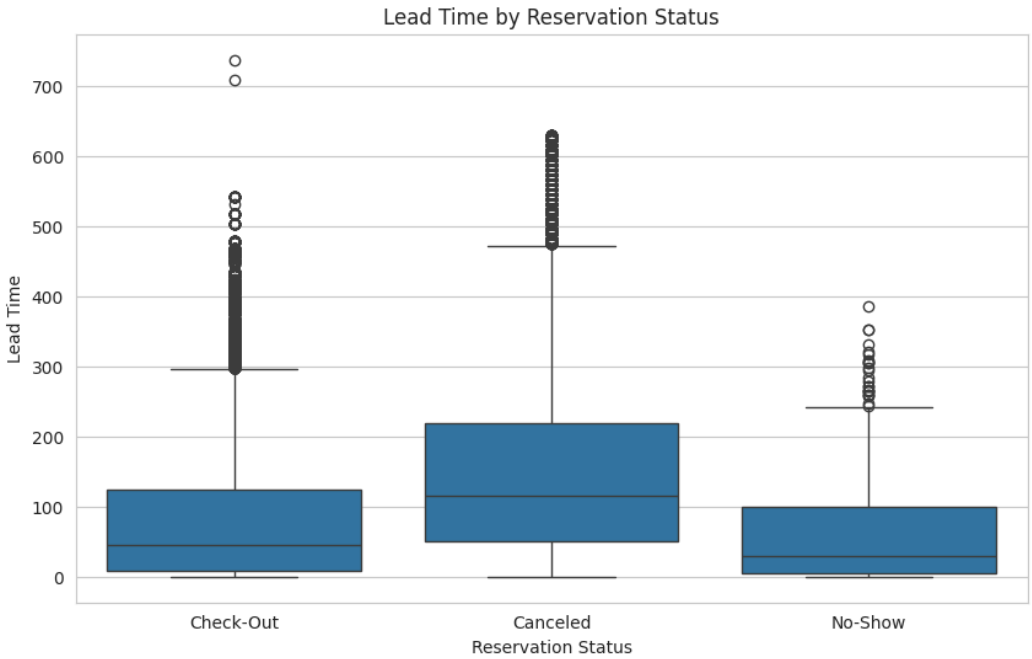


Figure 7: Lead Time by Reservation Status

Canceled bookings generally show longer lead times than completed stays. A longer interval between booking and arrival creates more opportunity for plans to change, which

helps explain the predictive value of lead_time.

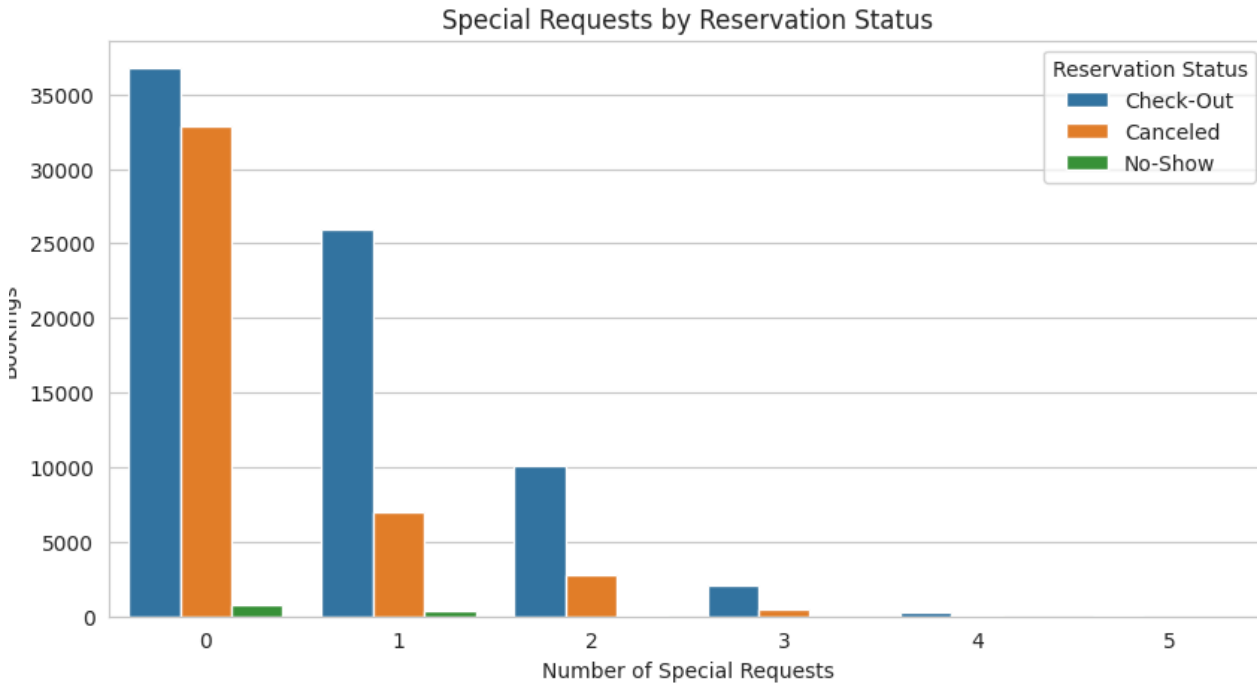


Figure 8: Special Requests by Reservation Status

Bookings with more special requests are more frequently associated with completed stays. Special requests may indicate stronger customer engagement or commitment to the reservation.

3 Dataset Pre-processing

Preprocessing was performed as a leakage-safe pipeline. Each data-quality problem is discussed first, followed by the solution and justification, as required by the project instructions.

3.1 Problem 1: Null and Missing Values

Four columns contained missing values: company (112,593), agent (16,340), country (488), children (4). The missingness was highly uneven, requiring different treatments rather than a single rule.

Table 3: Missing-Value Treatment

Feature	Missing	Solution	Reason
company	112,593	Drop column	94.31% missing
agent	16,340	Use 'Unknown'	Categorical identifier
country	488	Most-frequent imputation	Categorical feature
children	4	Median imputation	Numerical feature

The company column was deleted because approximately 94.31% of its entries were absent. Missing agent identifiers were treated as an Unknown category. Country and children were imputed insidethepreprocessing pipeline, whose parameters were fitted only on training data to prevent informationfromthetest set influencing training.

3.2 Problem 2: Categorical Variables

The dataset contains nominal text variables with no meaningful numerical order. OneHotEncoder was therefore applied with handle_unknown='ignore'. This produces a separate binary indicator for each observed category while ensuring that unseen test categories do not cause prediction failure. Three target labels were converted to integers with LabelEncoder.

3.3 Problem 3: Feature Scaling

Numerical features use very different units and ranges. For example, lead time may contain

hundreds of days, ADR is a monetary rate, while binary indicators contain only zero and one. StandardScaler was applied after median imputation. Scaling is especially important for distance-based KNN, gradient-based neural networks, logistic regression, and K-Means clustering.

3.4 Invalid Records, Leakage, and Feature Engineering

A total of 180 records with zero adults, zero children, and zero babies were removed because they don't represent valid guest bookings. The columns is_canceled, reservation_status_date, andassigned_room_type were removed to avoid direct or temporal leakage. Three derived features were then created: total_nights, total_guests, and is_family.

3.5 Feature Selection

After imputation and one-hot encoding, the supervised input contained 565 processed features. SelectKBest with the ANOVA F-test was fitted on training data and retained the 30 most informative features. This reduced computational cost while preserving strong signals such as lead time, previous cancellations, deposit type, market segment, country, hotel type, special requests, parking spaces, and selected agent categories.

4 Dataset Splitting

After removing invalid records and creating derived features, 119,210 observations remained. The dataset was divided using a stratified 80/20 train-test split with random_state=42. Stratification preserved the three-class distribution, which is particularly important for the rare No-Show class.

Table 4: Dataset Split Summary

Partition	Samples	Percentage
Training set	95,368	80%
Test set	23,842	20%
Total	119,210	100%

The preprocessing transformer and feature selector were fitted only on the training subset and then applied to the test subset. The neural network additionally reserved 20% of the training subset asaninternal validation set for early stopping.

5 Model Training and Testing

Four supervised classification models representing distance-based, linear, probabilistic, and nonlinear learning approaches were evaluated. This satisfies the requirement to implement a neural network and at least two additional models.

5.1 K-Nearest Neighbors

KNN classifies each test observation according to nearby training observations in the scaled featurespace. The implementation used KNeighborsClassifier with n_neighbors=5. It achieved 81.96% accuracy, 0.6032 Macro F1, and 0.8842 weighted multiclass AUC. KNN produced the best Macro F1 and balanced accuracy among the evaluated models.

5.2 Logistic Regression

Logistic Regression learns linear decision boundaries and estimates multiclass probabilities. The implementation used max_iter=2000 to allow convergence. It achieved 78.85%accuracy, 0.5069MacroF1, and 0.8666 AUC. Although it performed reasonably on the two majority classes, it failed to correctly identify No-Show observations.

5.3 Gaussian Naive Bayes

Gaussian Naive Bayes applies Bayes' theorem under conditional-independence and Gaussian-distribution assumptions. It achieved only 27.18% accuracy and 0.2935 Macro F1. Its No-Show recall was 0.97, but precision was only 0.01, meaning it classified too many observations as No-shows and produced a large number of false positives.

5.4 Neural Network

The multilayer perceptron contained three hidden layers with 128, 64, and 32 neurons, ReLUactivations, the Adam optimizer, a batch size of 256, and an initial learning rate of 0.001. Early stopping monitored a 20% validation subset and stopped training after 31 iterations when validation performance no longer improved.

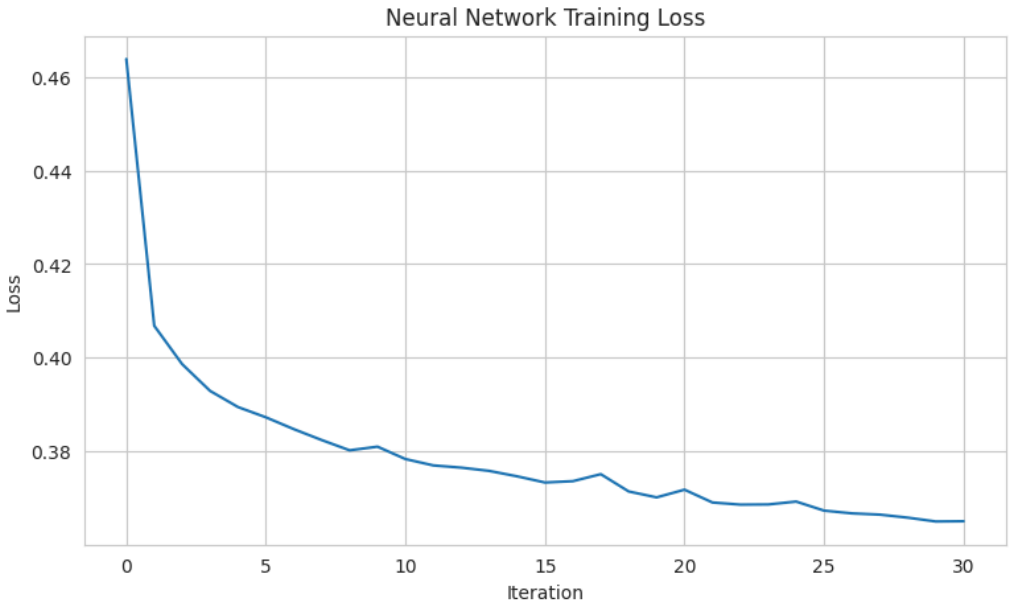


Figure 9: Neural Network Training Loss

Training loss decreased from approximately 0.464 to 0.365, while validation accuracy reached approximately 0.823. On the independent test set, the neural network achieved the highest accuracy(83.08%) and AUC (0.9098), although its Macro F1 was slightly lower than KNN because minority-class recall remained weak.

5.5 K-Means Clustering

For unsupervised analysis, reservation_status was removed before clustering. Numerical and categorical variables were independently imputed, scaled, and one-hot encoded. Truncated SVD reduced the 579-dimensional processed representation to 20 components while retaining 85.09% of the explained variance. The elbow method evaluated K values from 2 to 14 and selected K=7.

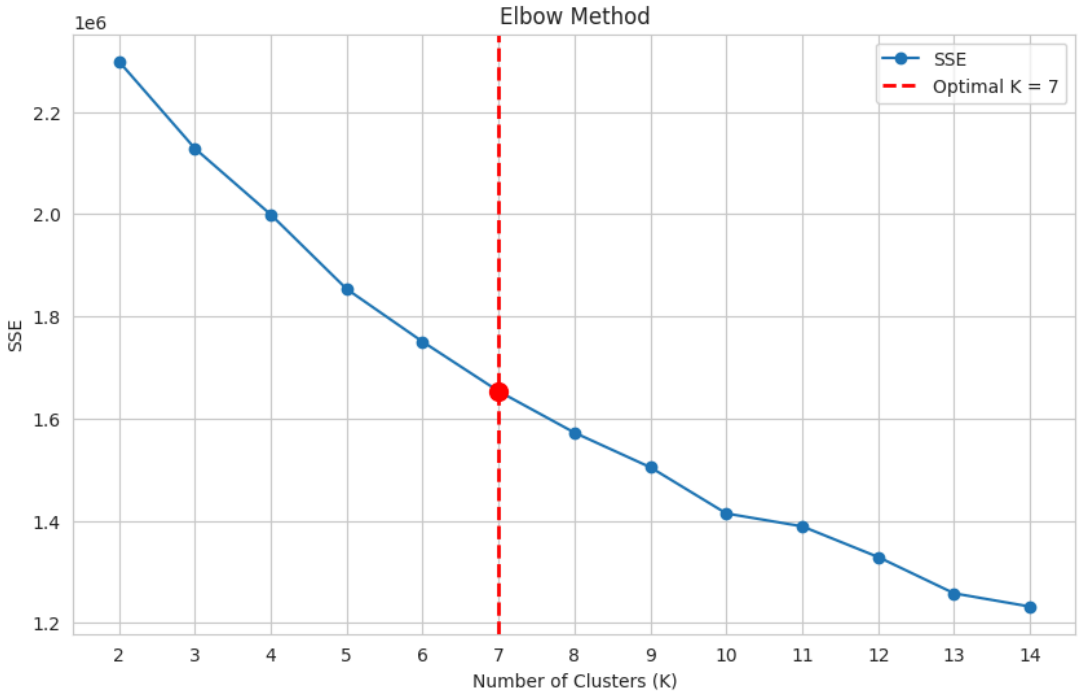


Figure 10: Elbow Method for Selecting the Number of Clusters

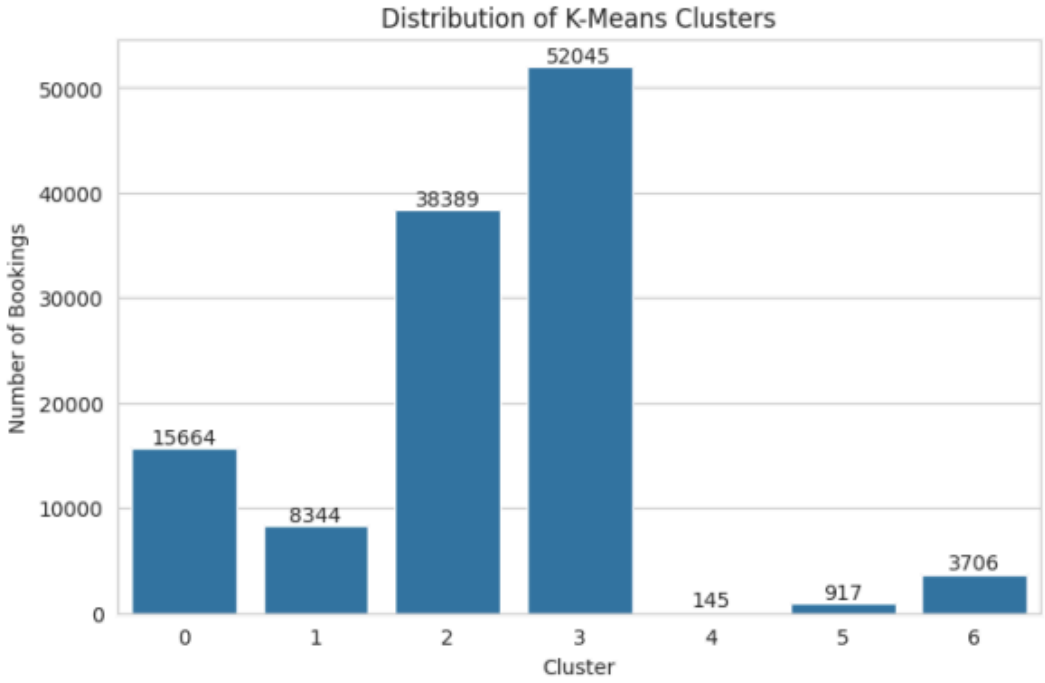


Figure 11: Distribution of the Seven K-Means Clusters

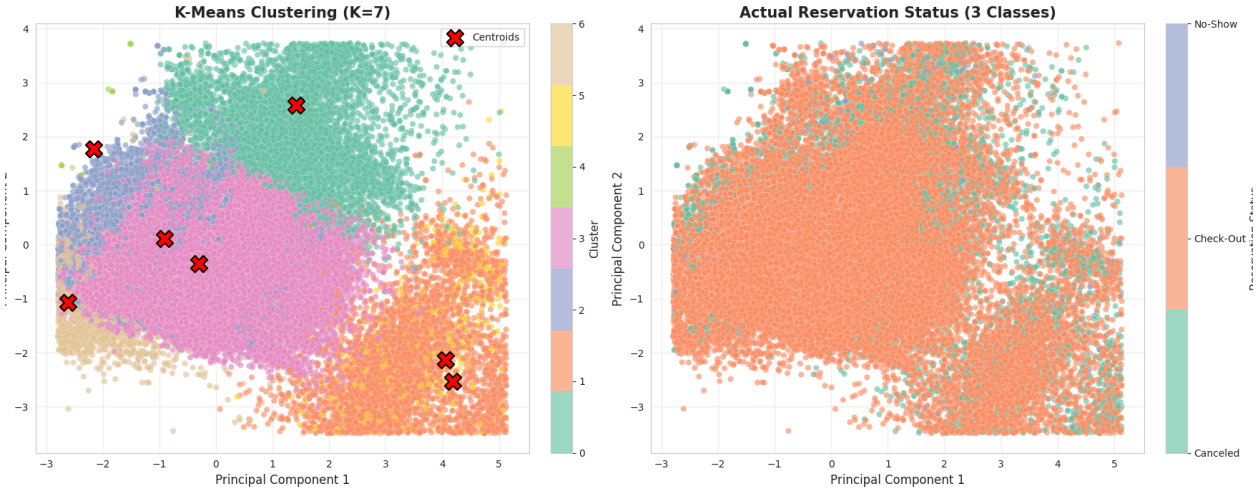


Figure 12: K-Means Clusters Compared with Actual Reservation Status in PCA Space

The first two principal components explain 26.30% of the reduced-space variance. The clusters do not correspond one-to-one with the three reservation outcomes, which is expected because K-Means discovers geometric booking profiles without access to status labels. Cluster profiling indicates differences in leadtime, ADR, number of nights, party size, previous cancellations, booking changes, and special requests.

6 Model Selection and Comparison Analysis

Model selection was based on multiple complementary measures rather than accuracy alone. This is essential because the extreme rarity of No-Show can cause majority-class performance to dominate weighted averages.

6.1 Accuracy Comparison

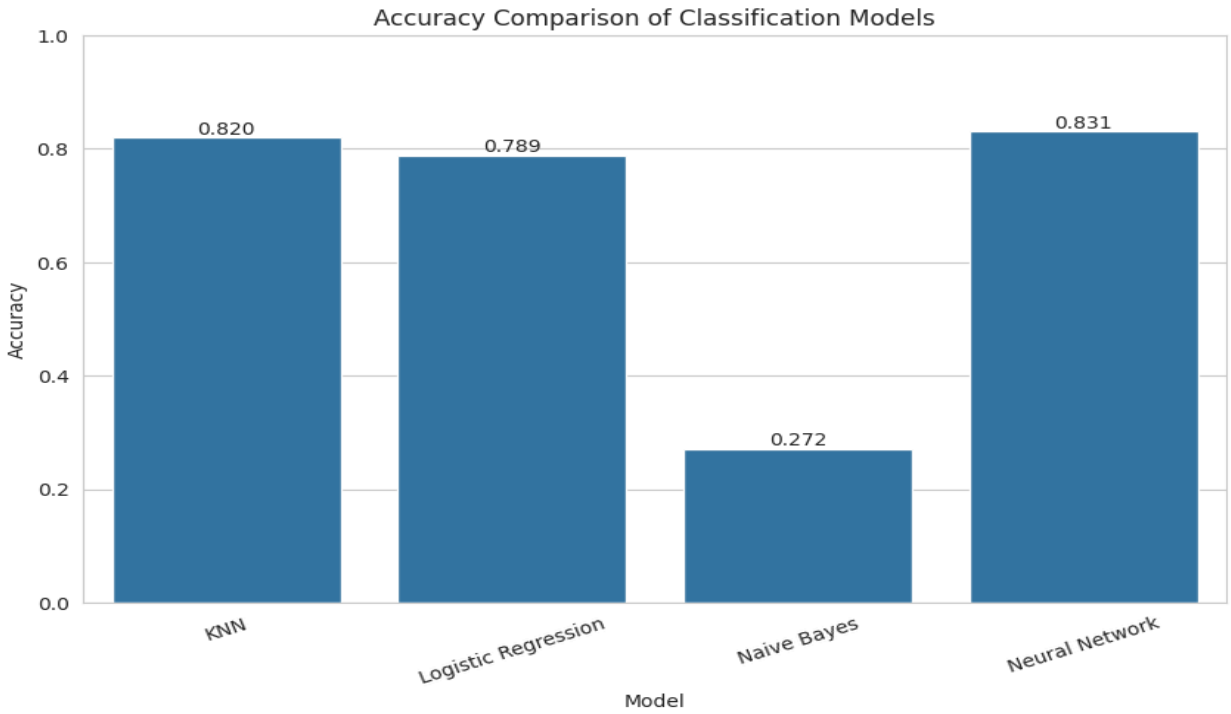


Figure 13: Prediction Accuracy Comparison Across All Models

The Neural Network achieved the highest accuracy at 83.08%, followed by KNN at 81.96%, Logistic Regression at 78.85%, and Naive Bayes at 27.18%.

6.2 Precision, Recall, and F1 Comparison

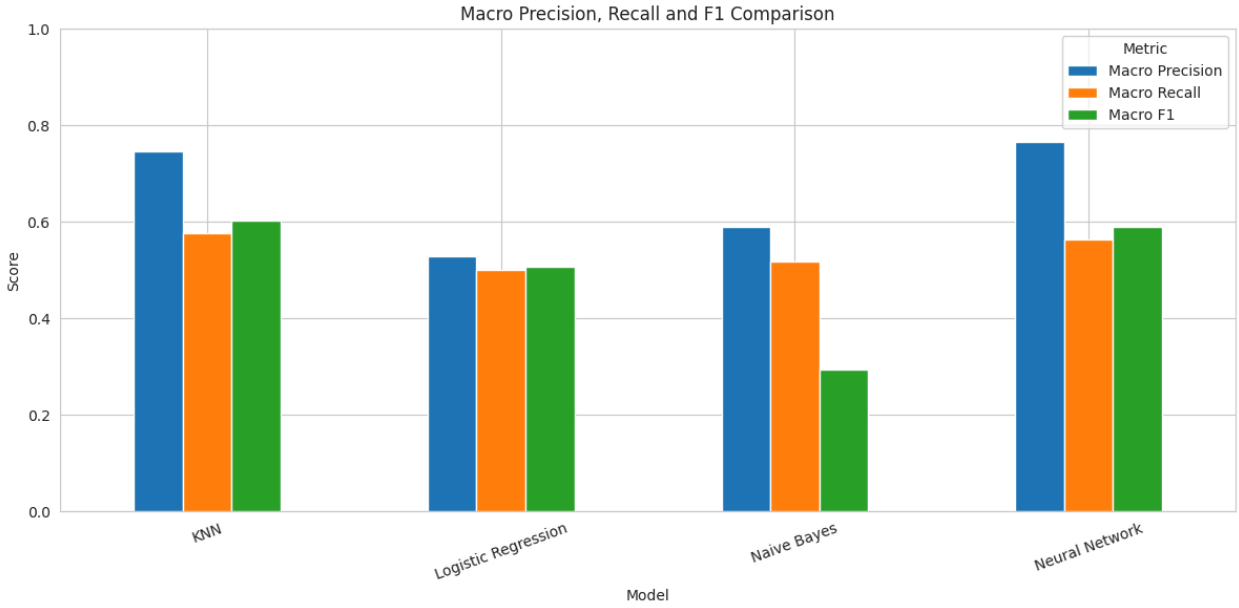


Figure 14: Macro Precision, Recall, and F1 Comparison

KNN achieved the highest Macro F1 (0.6032) and balanced accuracy (0.5760), indicating the best overall balance across the three classes. The Neural Network achieved the highest Macro Precision but lower Macro Recall than KNN. Weighted scores are higher than macro scores because performance is dominated by the two majority classes.

6.3 Confusion Matrices

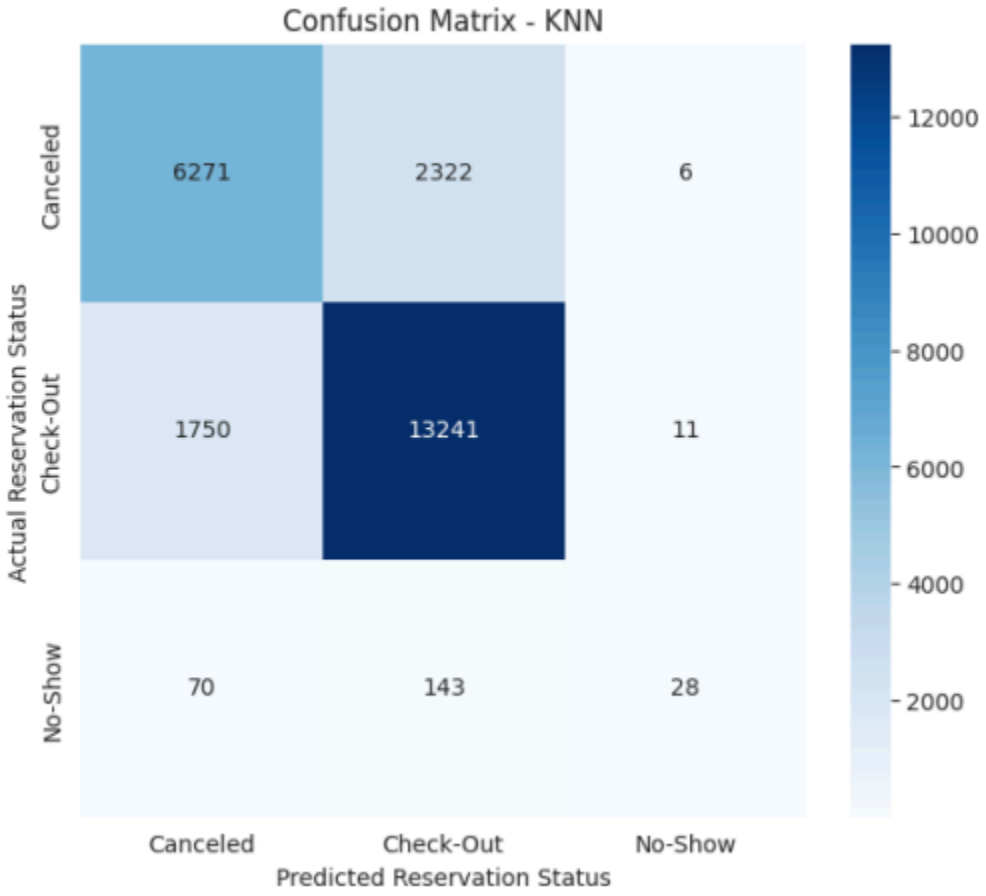


Figure 15: Confusion Matrix - KNN

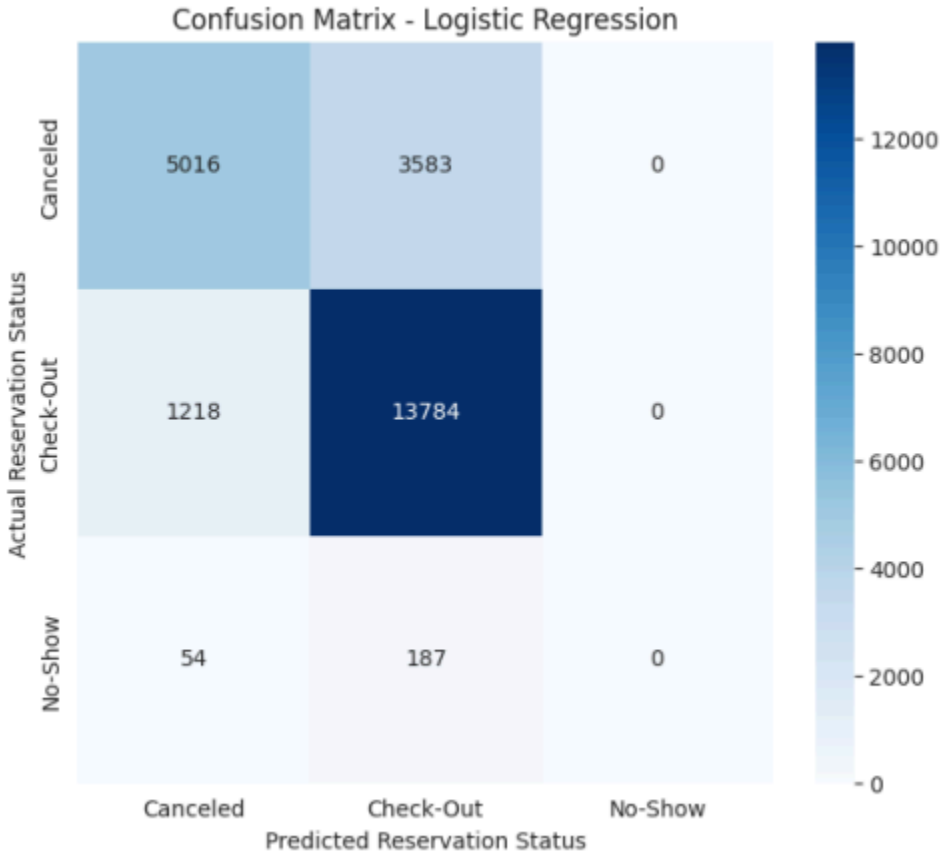


Figure 16: Confusion Matrix - Logistic Regression

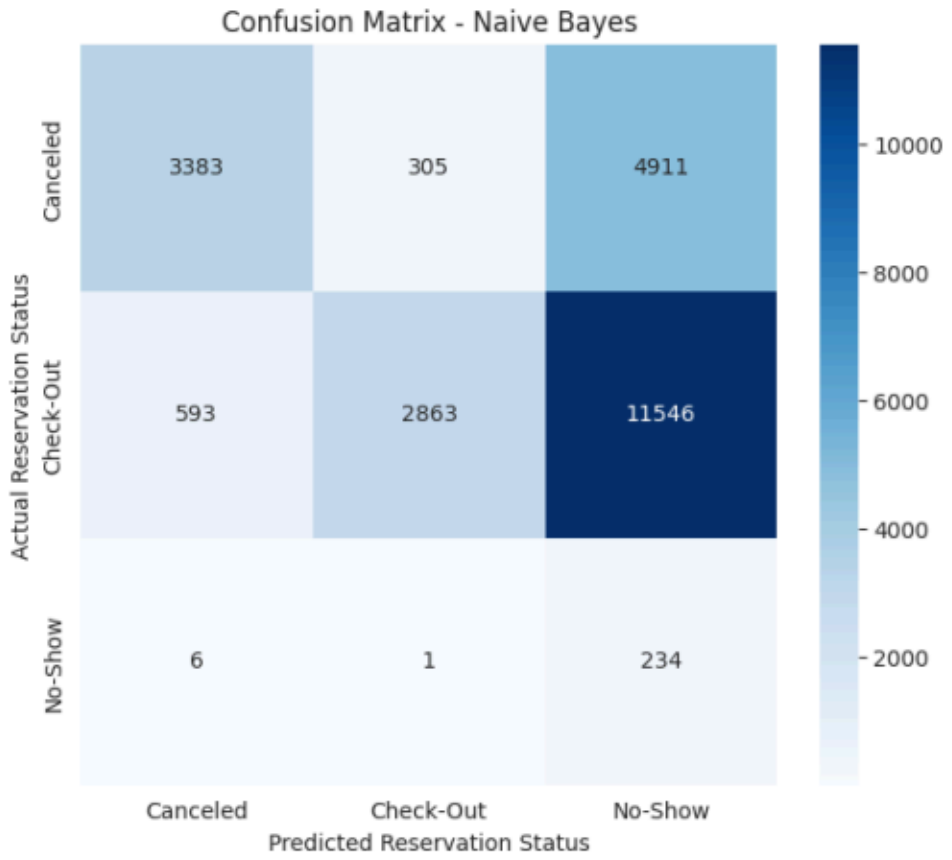


Figure 17: Confusion Matrix - Naive Bayes

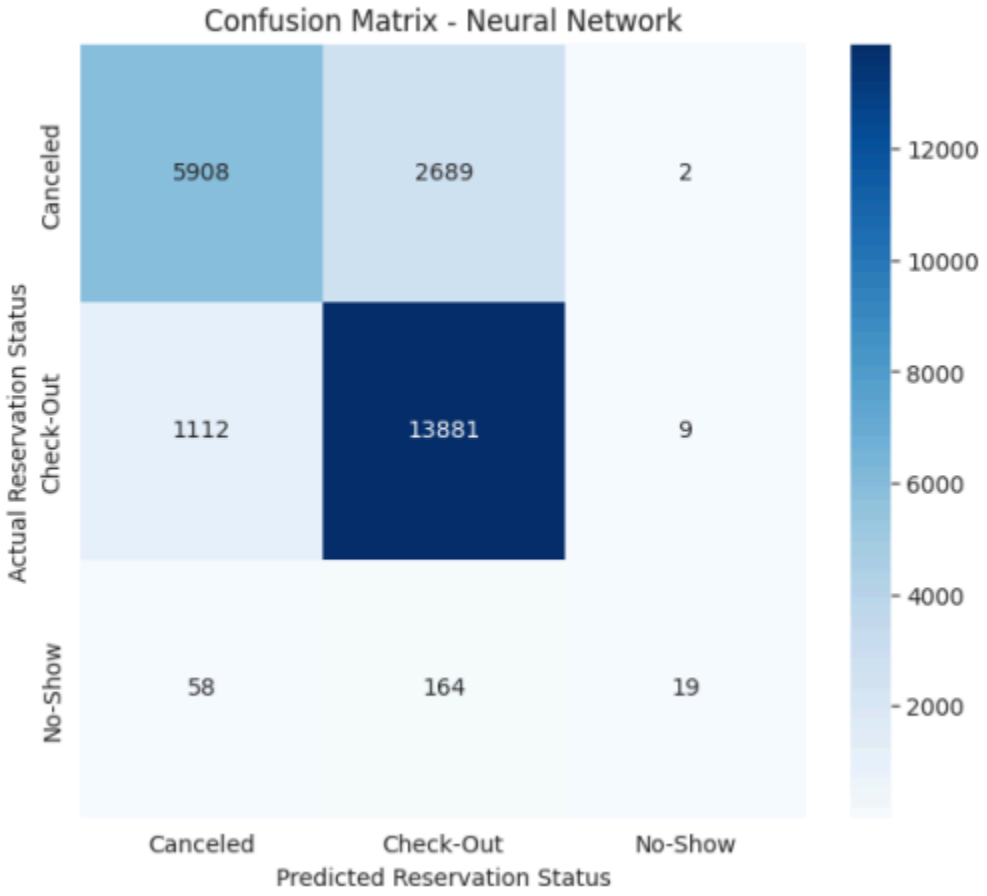


Figure 18: Confusion Matrix - Neural Network

The confusion matrices confirm that KNN and the Neural Network classify the two majority classes relatively well but identify only a small portion of no-shows. Logistic Regression identifies no no-shows correctly. Naive Bayes detects most no-shows but incorrectly assigns the No-Show label to many canceled and completed bookings.

6.4 AUC and ROC Curves

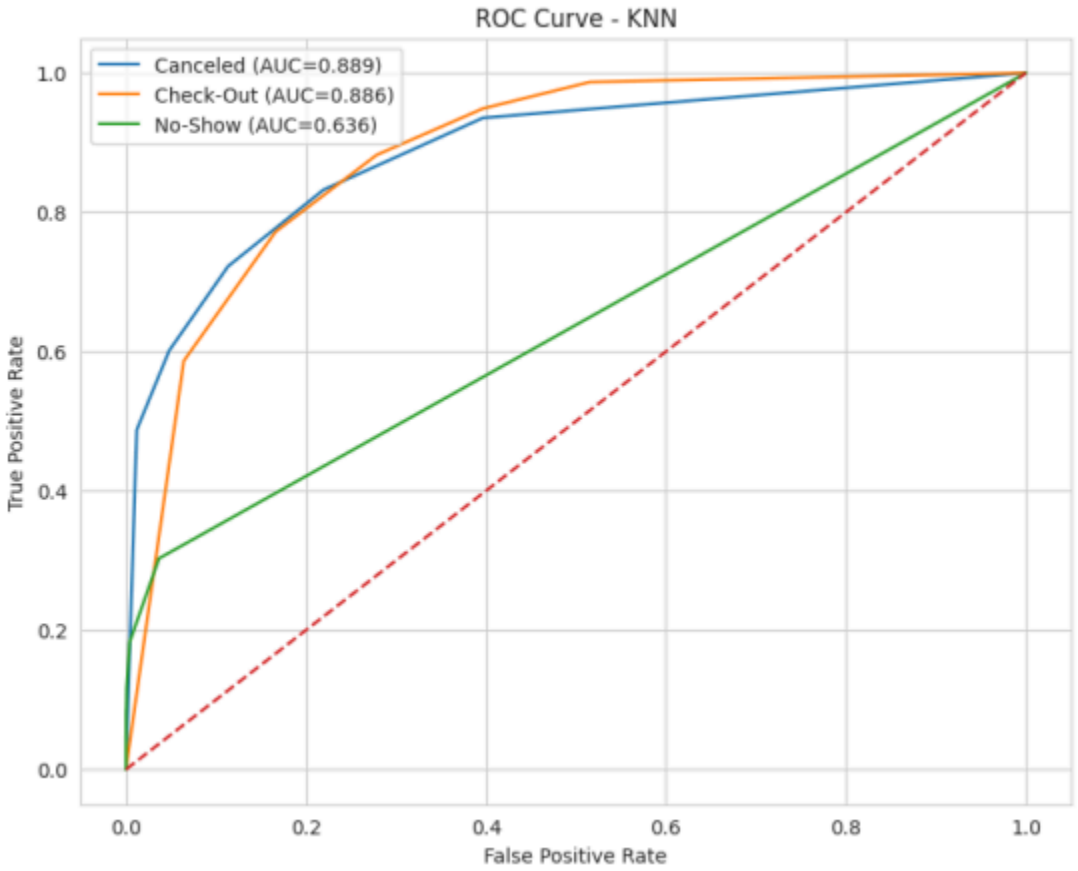


Figure 19: Multiclass ROC Curves - KNN

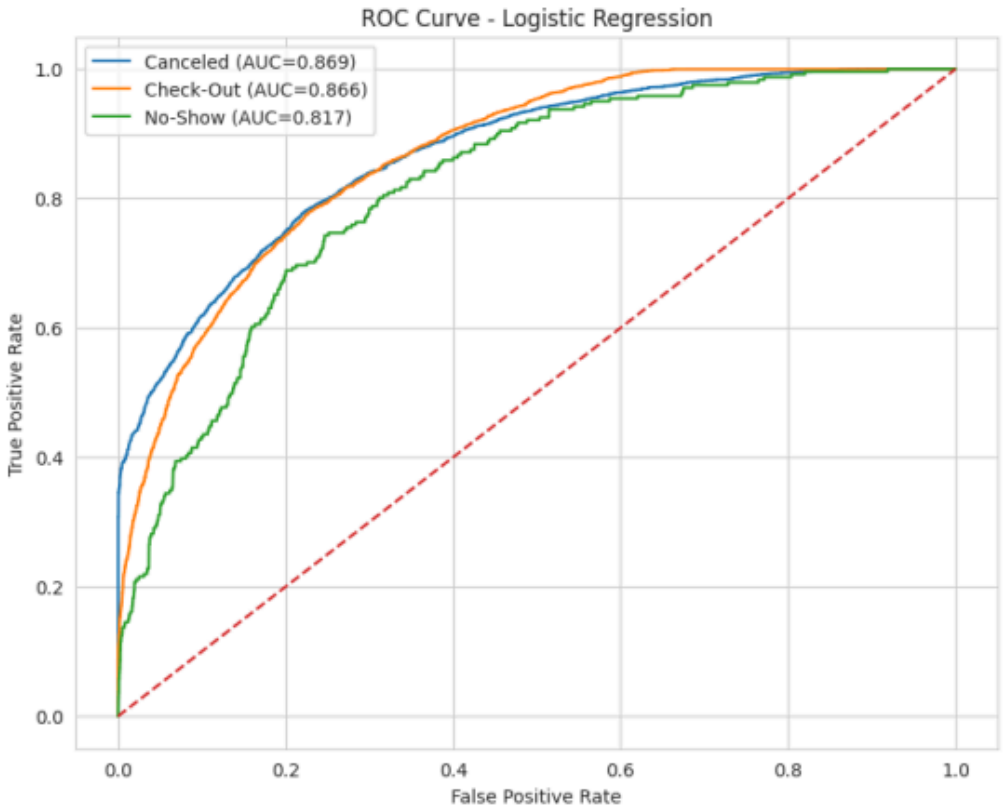


Figure 20: Multiclass ROC Curves - Logistic Regression

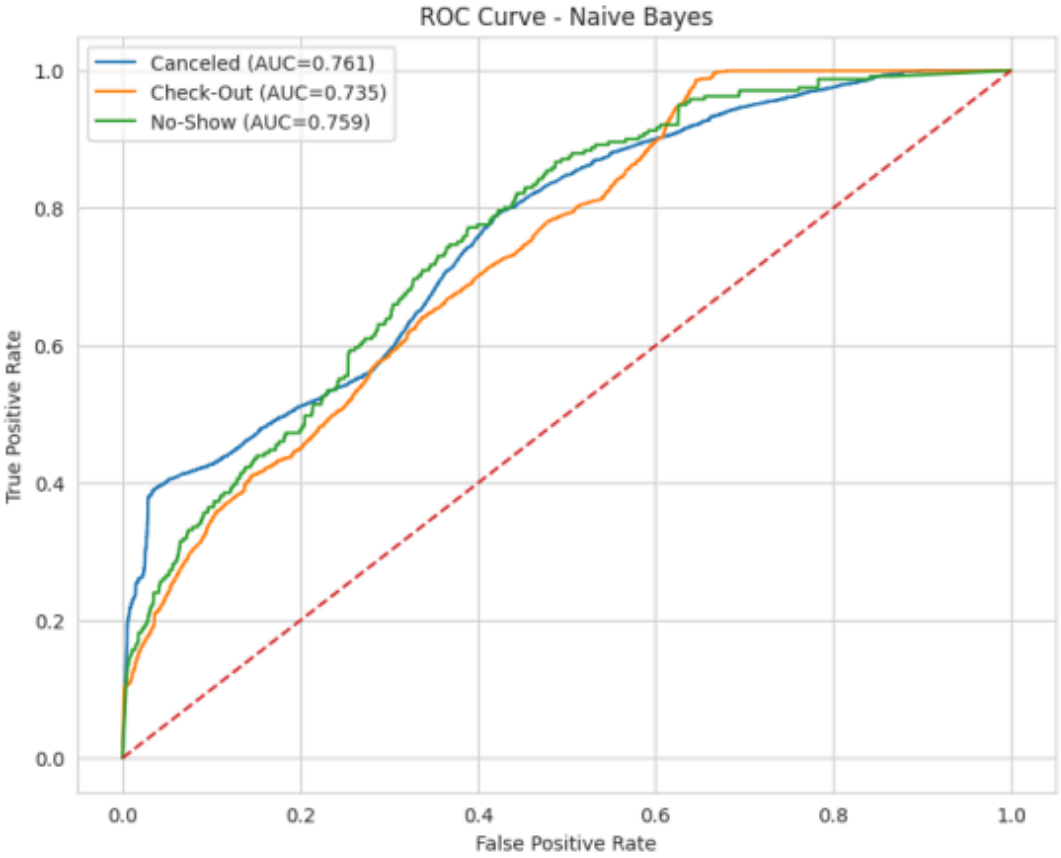


Figure 21: Multiclass ROC Curves - Naive Bayes

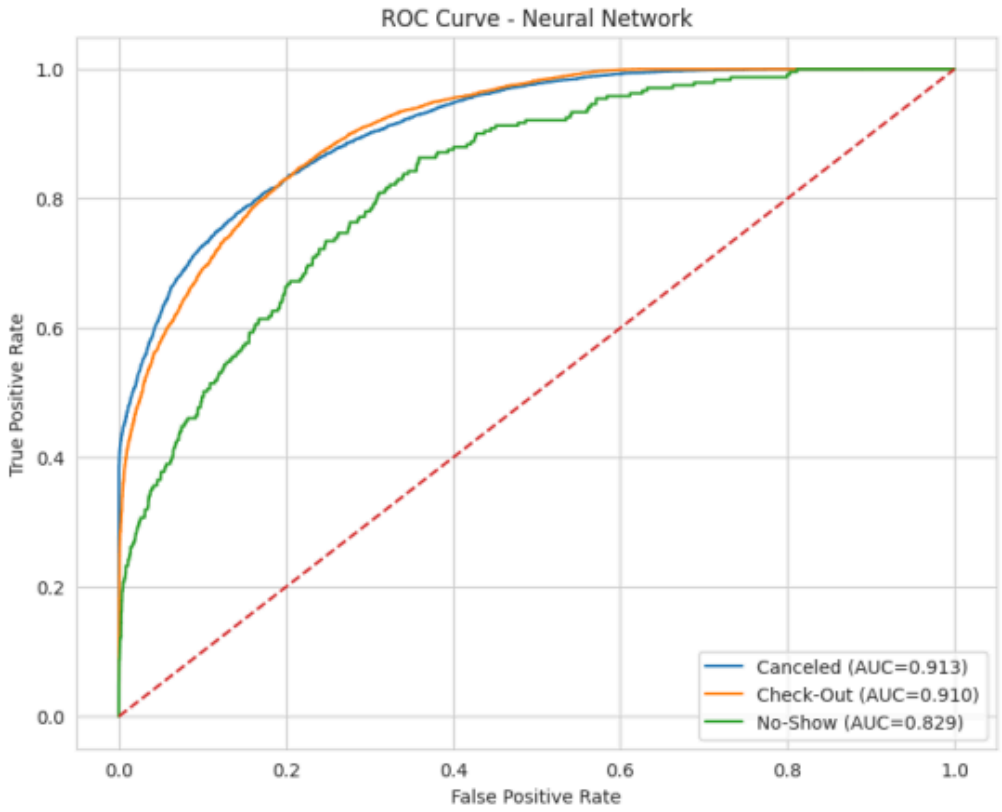


Figure 22: Multiclass ROC Curves - Neural Network

The ROC curves were computed using a one-vs-rest multiclass formulation. The Neural Network achieved the highest weighted AUC (0.9098), followed by KNN (0.8842), Logistic Regression (0.8666), and NaiveBayes (0.7447). AUC measures ranking and discrimination across thresholds and should be interpreted together with class-specific precision and recall.

6.5 Comprehensive Results Comparison

Table 5: Comprehensive Model Performance Comparison

Model	Accuracy	Balanced Acc.	Macro F1	Weighted AUC
KNN	0.8196	0.5760	0.6032	0.8842
Logistic Regression	0.7885	0.5007	0.5069	0.8666
Naive Bayes	0.2718	0.5184	0.2935	0.7447
Neural Network	0.8308	0.5637	0.5896	0.9098

No single model dominates every metric. The Neural Network is best by accuracy and weighted AUC, whereas KNN is best by Macro F1 and balanced accuracy. For a deployment that values performance across all classes, KNN is the more balanced choice among the tested models. For overall discrimination and majority-class accuracy, the Neural Network is stronger. Both require improvement before reliable No-Show prediction.

7 Conclusion

7.1 What Do We Understand from the Results?

The experiments demonstrate that hotel reservation outcomes contain learnable patterns. The Neural Network achieved 83.08% accuracy and 0.9098 AUC, while KNN achieved the strongest Macro F1 and balanced accuracy. These results suggest that both nonlinear interactions and local similarity patterns are useful for distinguishing completed and canceled reservations. Nevertheless, the rare No-Show class remains difficult to identify reliably.

7.2 Model Performance Comments

The Neural Network provided the highest overall accuracy and discrimination, but KNN produced the most balanced class-level result. Logistic Regression offered a simpler baseline but ignored the No-Show class. Gaussian Naive Bayes performed poorly because its independence and distribution assumptions don't fit the encoded booking features; its high No-Show recall came at the cost of extremely low precision and overall accuracy.

7.3 Why Are We Getting Such Results?

Several factors explain the observed results. Lead time, deposit type, previous cancellations, special requests, hotel type, market segment, and customer characteristics contain meaningful predictive information. Scaling supports distance- and gradient-based learning, while feature selection reduces dimensionality. The Neural Network can learn nonlinear interactions, and KNN can exploit local neighborhoods. However, because No-Show represents only about one percent of the data, optimization is naturally dominated by Check-Out and Canceled observations.

7.4 Challenges Faced

The principal challenges were severe class imbalance, large amounts of missing company and agent information, high-cardinality categories such as country and agent, prevention of target leakage, computational cost, interpretation of encoded categorical correlations, and selection of a meaningful number of K-Means clusters. Another challenge is selecting the best model when accuracy, AUC, balanced accuracy, and Macro F1 identify different winners.

7.5 Final Recommendations

Future work should collect more no-show cases, experiment with training-only oversampling or class-aware learning, tune model hyperparameters with cross-validation, and validate models chronologically using booking dates. Ensemble methods could also be evaluated. Operational deployment should monitor Macro F1, minority-class recall, calibration, and the business cost of false alerts rather than relying only on accuracy.